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Multimedia Systems

Simplified MPEG-4 Video Encoder Pipeline

Project Report

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1. Introduction

The system implements a simplified video compression pipeline inspired by MPEG standards. It processes a sequence of image frames and compresses them using both spatial and temporal redundancy reduction techniques. The compressed data is then encoded into a binary file and later reconstructed into a video.

I. Pipeline Description

2. Pre-processing Stage:

Each input frame is first read from the dataset and converted from the BGR color space to YCbCr. This transformation separates luminance (Y) from chrominance (Cb and Cr), allowing more efficient compression since the human eye is more sensitive to brightness than color.

After conversion, chrominance channels are downsampled using 4:2:0 subsampling to reduce spatial resolution and data size while preserving visual perception.

3. Intra-frame Coding (I-Frames)

Intra-frame compression is applied independently to key frames (I-frames). Each frame is divided into 8×8 blocks. For each block:

- A level shift is applied (-128)
- Discrete Cosine Transform (DCT) converts spatial data into frequency domain
- Coefficients are quantized using a JPEG-like quantization matrix

This process reduces high-frequency details that are less perceptible to the human eye. At the decoder, inverse quantization and inverse DCT are applied to reconstruct the frame.

4. Inter-frame Coding (P-Frames)

Temporal redundancy is exploited using motion estimation. Each P-frame is divided into 16×16 macroblocks.

For each macroblock:

- A search is performed in the previous reconstructed frame within a defined search window
- The best matching block is selected using Sum of Absolute Differences (SAD)
- Motion vectors are stored

A predicted frame is generated using these motion vectors. The residual (difference between original and predicted frame) is then encoded using the same DCT-based compression method as I-frames

5. Entropy Coding

All compressed data (DCT coefficients, residuals, motion vectors, and frame metadata) is serialized and compressed using a lossless compression algorithm (zlib). The result is stored in a single `.bin` file, significantly reducing storage size.:

6. Decoding Process

The decoding process reverses all previous steps:

- The binary file is decompressed
- I-frames are reconstructed using inverse DCT
- P-frames are reconstructed using motion compensation and residual addition
- Frames are combined to reconstruct the full video sequence

7. Output

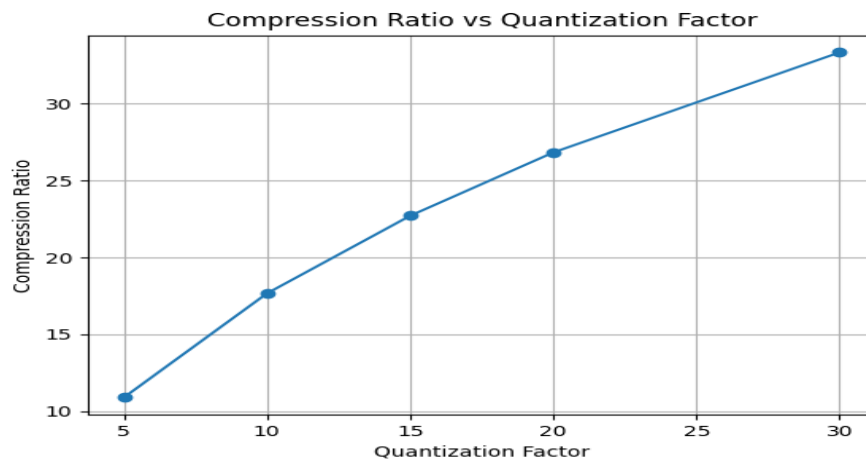
The final output of the system is a reconstructed video that approximates the original input sequence, demonstrating successful compression and decompression using spatial and temporal redundancy reduction techniques.

II. Design Choices & Justification

- **YCbCr color space** is used because it separates luminance (Y) from chrominance (Cb, Cr), matching human vision sensitivity and improving compression efficiency.
- **4:2:0 chroma subsampling** reduces Cb and Cr resolution since the human eye is less sensitive to color details than brightness.
- **8×8 DCT blocks** are chosen because they are standard in JPEG, offering a good balance between compression efficiency and computational cost.
- **Quantization matrix (JPEG-like)** reduces high-frequency coefficients more strongly, removing visually less important details and increasing compression.
- **16×16 macroblocks for motion estimation** provide a balance between accuracy and computational complexity in temporal compression.
- **SAD (Sum of Absolute Differences)** is used for block matching because it is simple and computationally efficient.
- **Residual coding for P-frames** improves compression by encoding only prediction errors instead of full frames.
- **zlib entropy coding** is applied as a final lossless compression stage to further reduce file size.
- **GOP structure (I and P frames)** ensures reference frames for reconstruction while maximizing compression efficiency.

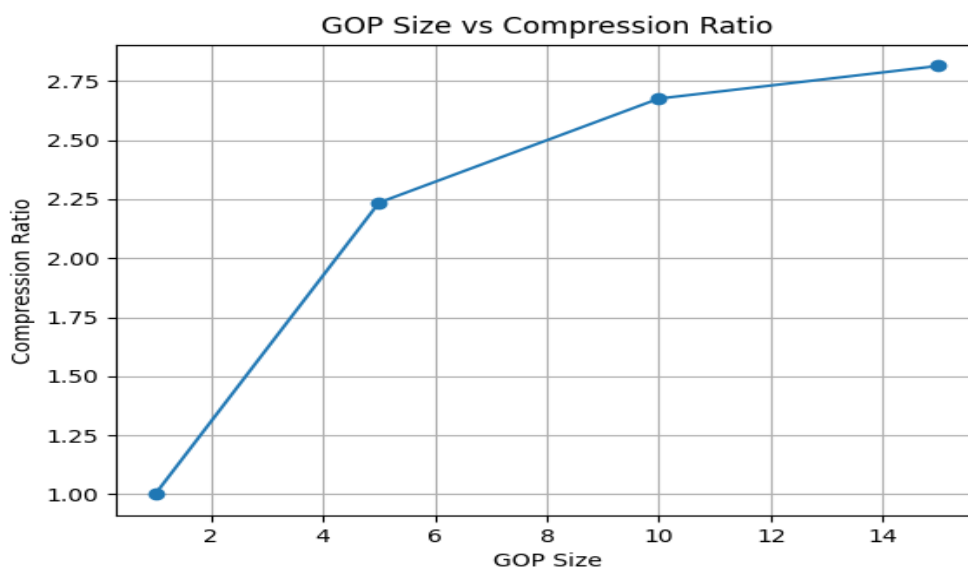
III. Experimental Analysis

- Compression Ratio vs Quantization Factor :



Increasing the quantization factor improves the compression ratio because more DCT coefficients, especially high-frequency ones, become zero after quantization. However, stronger quantization also reduces reconstruction quality due to information loss.

- Effect of GOP Size on Compression:



The experiments show that increasing GOP size generally improves compression ratio. However, very large GOP values may increase prediction errors and reduce reconstruction quality due to error propagation.